Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 24 GB of memory. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the training loss.

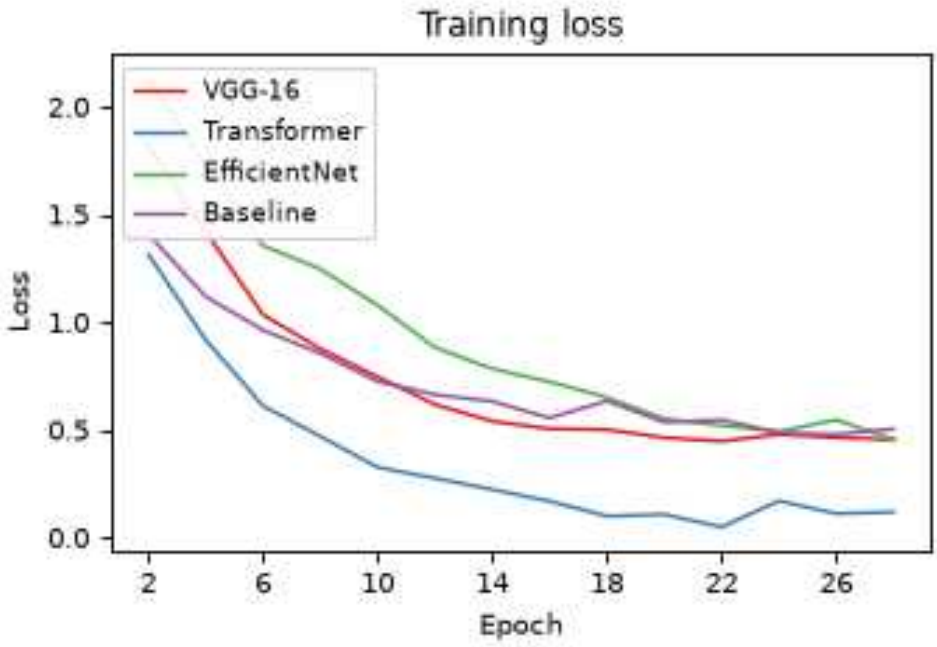


Figure 1: Training loss for the models in this experiment.

Transformer has a loss of 0.167 at epoch 16. At epoch 2, the loss of VGG-16 is 1.863 (Figure 1). VGG-16 has a loss of 0.500 at epoch 18. Transformer has a loss of 0.11 at epoch 26.

Figure 2 shows the test accuracy during training.

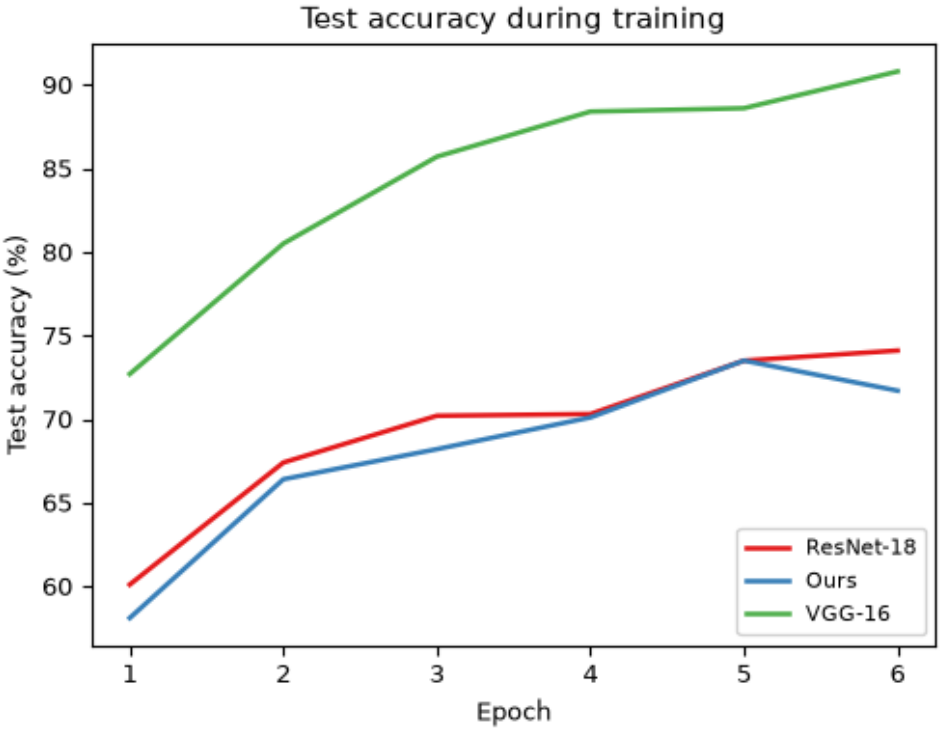


Figure 2: Test accuracy during training for the models in this experiment.

Ours reaches about 60% test accuracy at epoch 1. Ours reaches 74.1% test accuracy at epoch 6. ResNet-18 attains a test accuracy of 88.4% at epoch 4 (Figure 2). As shown in Figure 2, Ours attains a test accuracy of 66.3% at epoch 5.

3 Discussion

SVM reaches 89.2% accuracy on STL-10. We note that the training was stable for all models.